

Ann Song

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Education

Choate Rosemary Hall

Expected May 2025

4.11 UW, 4.49 W GPA

Deans' List for Choate Rosemary Hall Students Who Excel in All Subjects (All Terms)

Choate Rosemary Hall Excellence in Programming Award (May 2024)

Work and Volunteer Experience

Computer Science Instructor @ *KTByte Computer Science Academy*

October 2021 - Present

- Taught Java, Processing, and ACSL Contest Preparation to students 8 - 18 years old
- Collaborated with Teaching Assistants to provide students with the best experience possible
- Hosted office hours where students come in with questions on homework

Lead Instructor and Course Creator @ *Robotics and Beyond*

July 2023 - Present

- Created and taught Java curriculum at Robotics and Beyond
- Led and designated tasks to a group of TAs to assist students that were struggling
- Developed a Generative Art with Processing course as a stepping stone to Java

Club President @ *Choate GirlTech Club*

September 2021 - Present

- Volunteered and mentored local kids in STEM topics
- Collaborated and developed a personalized one-on-one mentorship program with the local public library
- Hosted bi-weekly meetings for Choate GirlTech Club members

Connecticut Youth Ambassador @ *Million Girls Moonshot Flight Crew*

February 2024 - Present

- Created video content that spreads awareness for women in STEM

Research Intern @ *Yale University O'Hern Group*

June 2024 - August 2024

- Collaborated w/ 2 interns to create SVR ML model to predict protein-protein interactions

Teaching Assistant and VEX IQ Coach @ *CT STEM Academy*

August 2022 - July 2023

- Taught Scratch to 90 kids at NASA Astro Camp
- Assisted instructors in Girl Powered STEM Summer camp
- Coached VEX IQ to prospective team members

Extracurriculars

Lead Programmer, Technician, Website Manager @ *Choate FRC Team (7407)*

August 2023 - Present

- Utilized skills in Python, PID, robot vision and more
- Used computer vision for accurate robot odometry to shoot game pieces
- Communicated with other subteams to streamline code development
- Mentored new members interested in joining the programming subteam
- Maintained the programming team's Standard Operating Procedures and FRC RobotPy Toolkit

Independent Researcher

June 2022 - Present

- Utilized Machine Learning techniques to assess cognitive states of the brain
- Developed and implemented a machine learning model to forecast the progression to Alzheimer's Disease
- Published research findings in the Journal of Student Research in May 2023

Chief Technology Officer and Project Director @ Youth Research Initiative **August 2023 - Present**

- Built a Python webscraper to collect data on regional and state science fairs
- Developed and maintained a comprehensive dataset encompassing all pertinent information related to these fairs
- Led a team to formulate a heuristic model for assessing science fair complexities and designing an intuitive UI and UX

Research Project Co-Leader @ Informatics & Computer Science Society **October 2022 - October 2023**

- Led a group of 10 researchers to collect data on app insecurities
- Developed a method to programmatically modify requests to expose flaws in apps
- Co-authored research paper <https://doi.org/10.48550/arXiv.2310.14137>

Head of Website Engagement @ The Choate News 118th Masthead **October 2022 - Present**

- Led a team of 4 to upload articles and maintained The Choate News Website
- Developed pages of a newly designed website in 2023

Honors

- **FIRST Robotics**
 - 3rd place Alliance ~ CT State Championships 2024
 - Excellence in Engineering Award ~ New England District Championships 2024
 - Quality Award ~ New England District Event 2024
 - Gracious Professionalism Award ~ New England District Event 2024
 - Semi-finalists and Innovation in Control Award ~ World Championships Hopper Division 2023
 - Division Winner, 4th place in New England ~ New England District Championships 2023
 - Gracious Professionalism Award ~ New England District Championships 2023
 - 3rd place Alliance, Industrial Design Award ~ New England District Event 2023
 - 2nd place Alliance, Excellence in Engineering Award ~ New England District Event 2023
- **Connecticut Science and Engineering Fair**
 - 3rd place in High School Computational Biology and Bioinformatics 2024
 - 5th place in High School Biotechnology 2024
 - 3rd place in High School Life Science 2023
 - Petit Family Foundation Women in Science & Engineering Awards - High School Finalist 2022
 - American Society of Safety Professionals Award 2022
 - IEEE Award, Connecticut Section 2022
- **Computer Science Awards**
 - National Honorable Mention Winner of the NCWIT Award for Aspirations in Computing 2024
 - Connecticut Winner of the NCWIT Award for Aspirations in Computing 2023 & 2024
 - 1st Place Girls Programming League Competitive Programming Challenge 2022

Scores

- SAT 1580
- AP Computer Science A 5

Publications

Song, Ann. "Using Machine Learning to Forecast Progression from Cognitively Normal to Alzheimer's Disease."
Journal of Student Research 12.4 (2023): 1-1